Trajectory Classification with LSTM for Robotic Peg-in-Hole

Master's Machine Learning
Project – 2025–2026

Group 1 - LOUNAS Gana | MAGOUO NOCHI

UE5 Apprentissage

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Project Objectives

- Build a trajectory classifier for a 7-DoF peg-in-hole task with **3 classes**.

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  - Analysis of **feature influence** on performance.

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- Why sequence learning: class information is encoded in **temporal dynamics**, not only static snapshots.
- Evaluation context: internal split (train/val/test), plus compatibility with an external test set.

Roadmap

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Dataset and Feature Blocks

- Files: `class1_trajectories.npy`, `class2_trajectories.npy`, `class3_trajectories.npy`

Block	**Meaning**	**Size**
`peg_pose`	peg position + quaternion	7
`peg_vel`	linear + angular velocity	6
`hole_pos`	hole position + quaternion	7
Total		**20**

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- Numerical stability: $\sigma = 0$ replaced by small ϵ.

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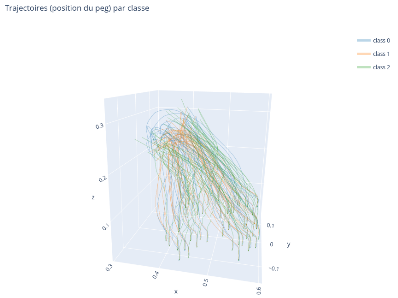
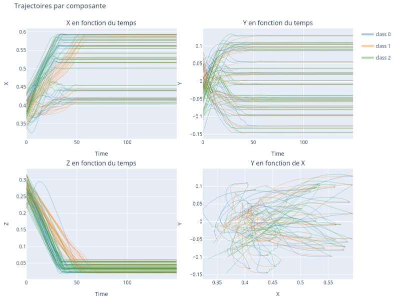
Trajectory Evidence Before Training





3D trajectories (x, y, z) by class. Projections: $x(t), y(t), z(t), y(x)$.

Class-dependent temporal patterns are visible before training, which motivates a temporal classifier.

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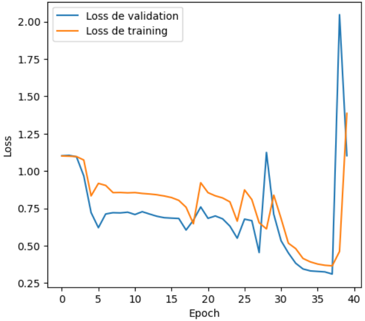
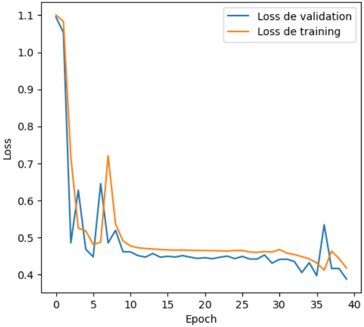
- Stability: gradient clipping $\|\nabla\|_{max} = 1.0$.

Practical Issues During Development

- **Reproducibility**: no fixed seed ⇒ difficult run-to-run comparison.



No normalization: noisier and less predictable convergence.



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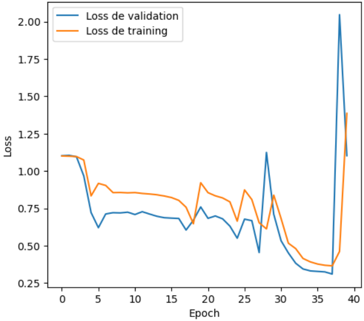
- **Optimization sensitivity**: preprocessing and regularization strongly alter convergence quality.



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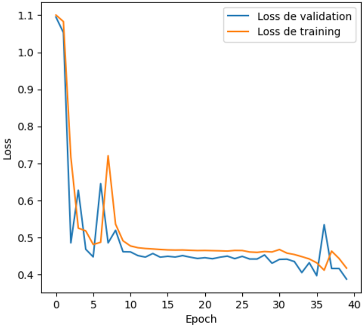

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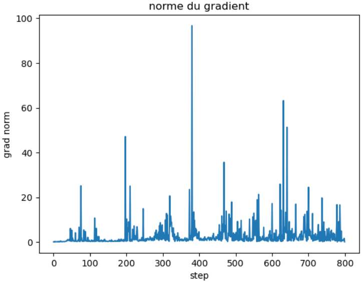
Exploding Gradients and Mitigation



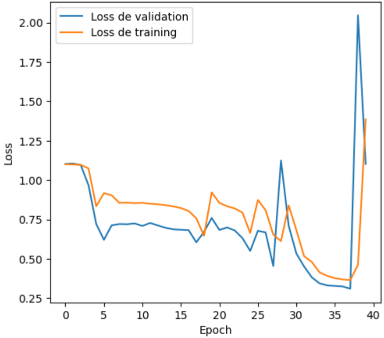
Gradient norm spikes observed during training.

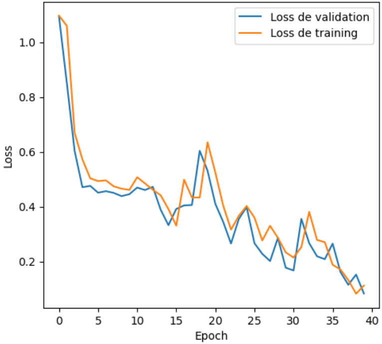
Observed issue
Large gradient spikes destabilize the parameter's updates.

Fix used
Gradient clipping in the training loop: `clip_grad_norm_(..., 1.0)`.

Effect
More stable updates and smoother final convergence.

Runs R1 and R2

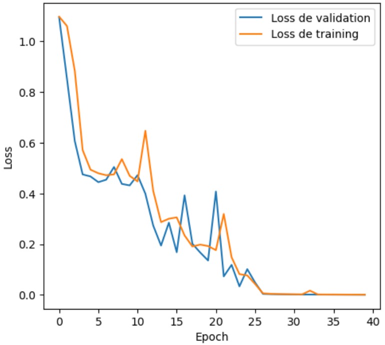




R1 baseline: loss diverges at the end — no normalization, no dropout, no clipping.

R2 (+z-score +dropout): **98.72%** test accuracy (77/78).

Run 3 : Best Run



R3 (+dropout + gradient clipping): **100%** test accuracy (78/78).

- Gradient clipping eliminates late-training spikes and ensures stable convergence.
- This configuration is selected as the final model.

Feature-Ablation Protocol

Subset	**Indices**	**Epoch budgets**
Peg pose + orientation	0:7	40, 80, 150
Peg linear+angular velocity	7:13	40, 80, 150
Hole pose + orientation	13:20	40, 150

- Goal: isolate which observation blocks truly carry discriminative information.

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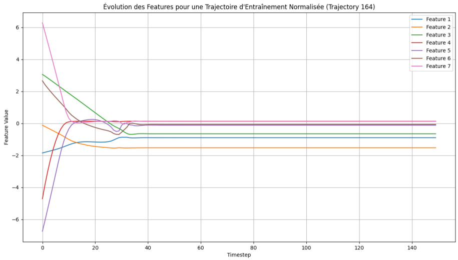
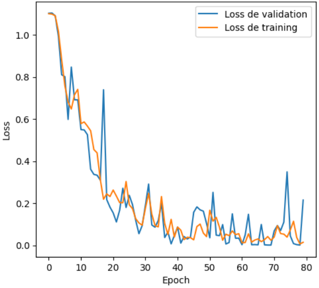
- Goal: isolate which observation blocks truly carry discriminative information.
- Same LSTM family and evaluation protocol; only feature subset and epoch budget vary.

Subset A: Peg Pose/Orientation Only (0:7)



Evolution of the 7 peg pose and orientation features.



Example loss curve (80 epochs).

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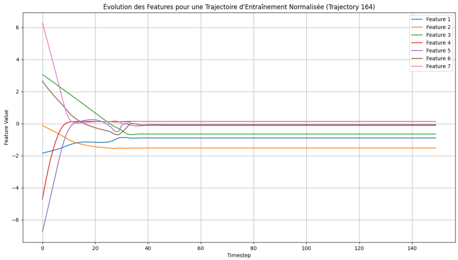

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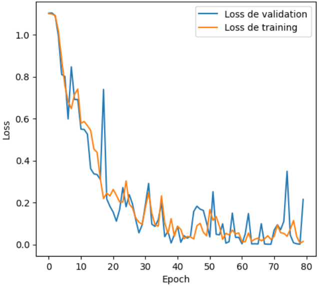
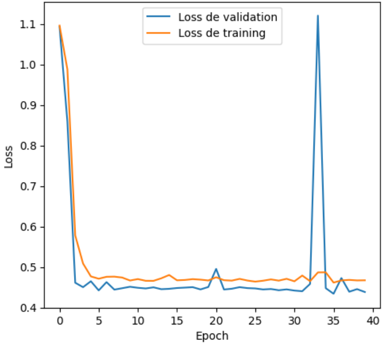


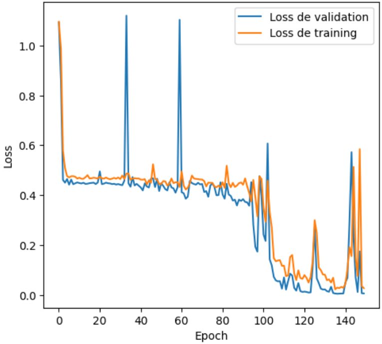
Example loss curve (80 epochs).

- Test performance: 98.72% (40 ep), 100% (80 ep), 100% (150 ep).
- Interpretation: The peg's pose and orientation alone already contain most class-discriminative signal.

Subset B: Peg Velocities Only (7:13)



40 epochs: moderate, unstable plateau.



150 epochs: late transition to low loss.

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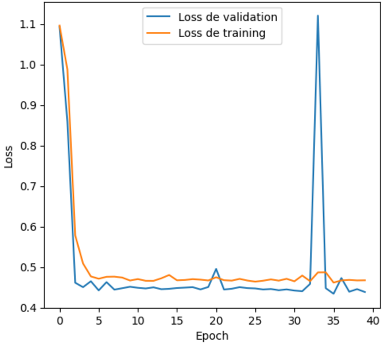

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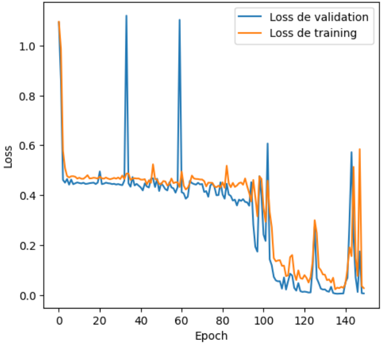

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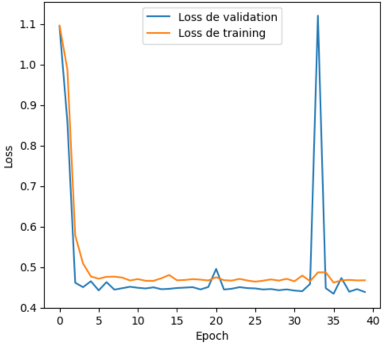

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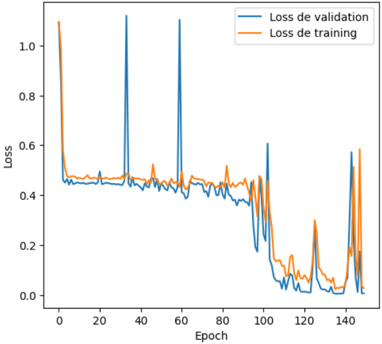

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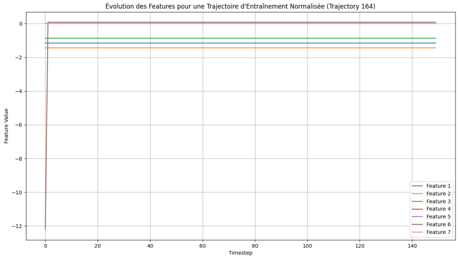
- Velocity is informative, but optimization is harder and slower.

Subset C: Hole Pose/Orientation Only (13:20)



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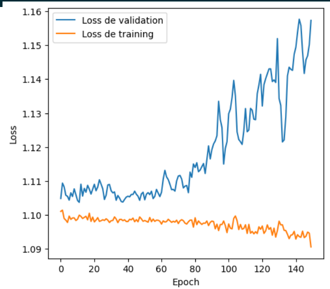

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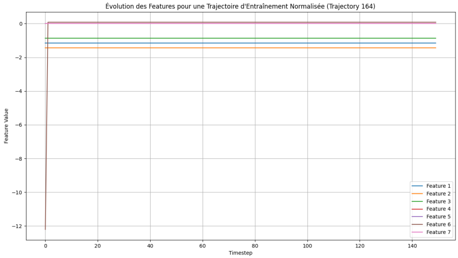

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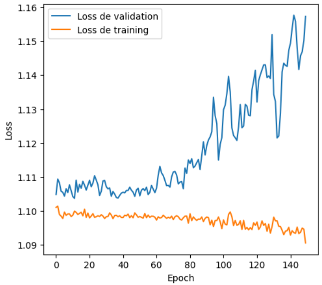

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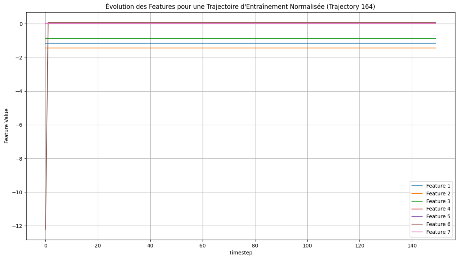

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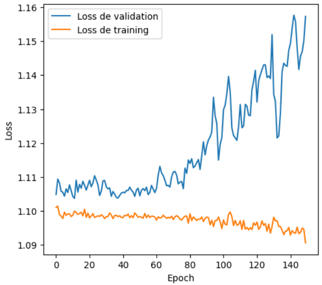

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- Interpretation: this subset is non-discriminative for trajectory classification.

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- Methodical ablation clarifies causality:
  - The peg's pose and orientation are highly informative,
  - The peg velocity is informative but optimization-sensitive,
  - The hole's pose/orientation alone is not discriminative here.
- Good accuracy comes from both **signal selection** and **stable optimization**, not from architecture choice alone.

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Backup: Why 2 LSTM Layers and Hidden Size 64?

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- A stronger baseline (hidden size 100) showed weaker behavior in this project context.
- With 768 trajectories and 150 timesteps, over-capacity increases instability/overfitting risk.
- Two recurrent layers keep modeling depth while staying computationally light.

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Backup: Model Size and Complexity

Component	**Shape/setting**	**Parameters**
LSTM layer 1	input 20, hidden 64	22,016
LSTM layer 2	input 64, hidden 64	33,280
Classifier head	`Linear(64,3)`	195
Total		**55,491**

- The model is compact ($\sim$ 55k params), which is appropriate for this dataset size.

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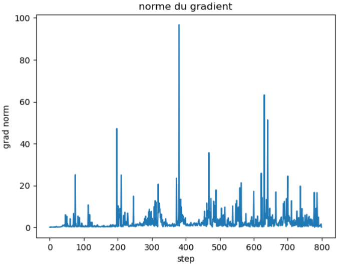
- Capacity is sufficient to fit complex temporal patterns without requiring very large-scale tuning.

Backup: Why Gradient Clipping at 1.0?



Observed gradient norm spikes before stabilization.

Rule used
`clip_grad_norm_(..., 1.0)`.

Reasoning
- recurrent training showed large norm spikes
- only large gradients are rescaled

Observed effect
stable late-epoch behavior and no final divergence.

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- **Hole pose/orientation (13:20)**: near-constant in these trajectories, so weak discriminative power alone.
- Ablation behavior is therefore physically coherent with the task dynamics.

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- Final external hidden test remains the decisive out-of-distribution validation.

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- Internal test set size is 78 trajectories: strong result, but still finite-sample uncertainty.
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- Single-seed reporting is reproducible but not a full multi-seed robustness study.
- Final external hidden test remains the decisive out-of-distribution validation.
- Data leakage risk was mitigated by train-only normalization statistics.

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- **Engineering tradeoff**: LSTM gave high accuracy with interpretable/stable training controls.
- GRU/Transformer remain valid extension baselines for future comparison.

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Run	**Test accuracy**	**Main effect**
R1	diverges	no regularization or normalization
R2	98.72% (77/78)	z-score + dropout improve generalization
R3	100.00% (78/78)	clipping stabilizes training

- Most impactful additions in this pipeline: **normalization + dropout + clipping**.

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- Most impactful additions in this pipeline: **normalization + dropout + clipping**.
- Internal split performance is excellent; external hidden test remains the final robustness check.

Backup: LSTM

- Reminder: vanilla RNNs are sensitive to vanishing/exploding gradients through BPTT.

$$i_t, f_t, o_t = \sigma(\cdot), \quad \tilde{c}_t = \tanh(\cdot), \quad c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$$

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- This architecture matches the project setting: long trajectories (150 steps) with dynamic class signatures.